

For problem 7, look at the example in the notes that starts off:

Example: calculus on power series

Find a power series representation for $\ln(x + 1)$ and state the interval of convergence.

Since we really only know the series $\frac{1}{1-x}$ at the moment, let's try to work with that. The connection to $\ln(x + 1)$ is that

$$\int \frac{1}{1+x} dx = \ln(1+x) + C,$$

so we want to replace x with $-x$ in $\frac{1}{1-x}$ and integrate the series term-by-term. We have

Here, we have a power series

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} (-x)^n$$

which converges on $(-1, 1)$ but diverges at the endpoints. However, when we take the integral

$$\int \frac{1}{1-x} dx = \int \sum_{n=0}^{\infty} \frac{(-x)^n}{n} = \sum_{n=0}^{\infty} -\frac{(-x)^{n+1}}{n+1}$$

and plug in $x = 1$, we get an alternating harmonic series, which converges.

For problem 6, though, I am having trouble coming up with an example of a power series that diverges at its endpoints but whose derivative converges there.
